Syntax and Semantics

Mathematical Preliminaries & Notation

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in this Lecture

- Sets (notation and basic definitions)
- Sequences and tuples
- Relations
- Functions
- Boolean logic and logical notation
- Types of proofs (by *construction*, *contradiction*, *induction*)

Sets

- a *set is a group of objects* (eg. numbers, symbols, other sets, ...)
- the objects in a set are called *elements* or *members*
- several descriptions

no repetitions + order is not relevant

$\in$ and $\notin$ denote set membership

  - listing of their elements: $\{7, 21, 57\}$, $\{x, y, z, \ldots\}$,
  - by rule of membership: $\{n \mid n = 2m \text{ for some } m \in \mathbb{N}\}$

Notable sets

$\emptyset = \{\}$ (*empty set* -the set with no elements)

$\mathbb{N} = \{0, 1, 2, 3, \ldots\}$ (set of *natural numbers*)

$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$ (set of *integers*)

$\mathbb{Q} = \left\{\frac{n}{m} \mid \text{for some } n, m \in \mathbb{Z} \text{ and } m \neq 0\right\}$ (set of *rationals*)

Sets (basic definitions)

For two generic sets A and B, we say that

- ***A is a subset of B***, written $A \subseteq B$, if every member of A is a member of B; (hence, $A = B$ iff $A \subseteq B$ and $B \subseteq A$).
- ***A is a proper subset of B***, written $A \subset B$, if $A \subseteq B$ and $A \neq B$;
- the ***union of A and B***, written $A \cup B$, is the smallest (w.r.t. subset inclusion) set that contains all the elements of A and B;
- the ***intersection of A and B***, written $A \cap B$, is the biggest (w.r.t. subset inclusion) set that is contained both in A and B;
- the ***complement of A***, written $\overline{A}$, is the set of all elements under consideration that are not in A;
- the ***powerset of A***, written $\mathscr{P}(A)$, is the set of all subsets of A.

Sequences & Tuples

- a *sequence* is an ordered list of objects
- we write them by listing their elements in between parenthesis

repetitions and order of objects matters!

$$(7,7,21,57)$$

- Sequences my be finite or infinite:
  - a finite sequence with k elements is called *k-tuple* (or simply *tuple*); a 2-tuple is also called *pair*;
  - infinite sequences are called *streams*.

Cartesian Product

Sets containing tuples are central to mathematics…

Definition

For two sets A and B, the ***Cartesian product*** (or ***cross product***) of A and B, written $A \times B$, is the set of all pairs of the form (a, b), where $a \in A$ and $b \in B$.

- **Example:** let $A = \{1,2\}$ and $B = \{x, y, z\}$, then
$$A \times B = \{(1,x), (1,y), (1,z), (2,x), (2,y), (2,z)\}.$$
- The ***k-fold Cartesian product*** of a set A is $A^k = \overbrace{A \times \ldots \times A}^{k-times}$; it contains k-tuples of elements of A.

Relations

- A ***k-ary relation*** (or simply ***relation***) between $A_1, \ldots, A_k$ is a set R containing k-tuples of elements of $A_1 \times \ldots \times A_k$;
- a relation between one set A, is called ***relation on A***

Example 1: for $A = \{x, y, z\}$ the following is a 2-ary relation

$$R = \{(x, x), (x, y), (y, z)\};$$

Example 2: "*less than or equal*" $\leq \subseteq \mathbb{N} \times \mathbb{N}$ is a 2-ary relation

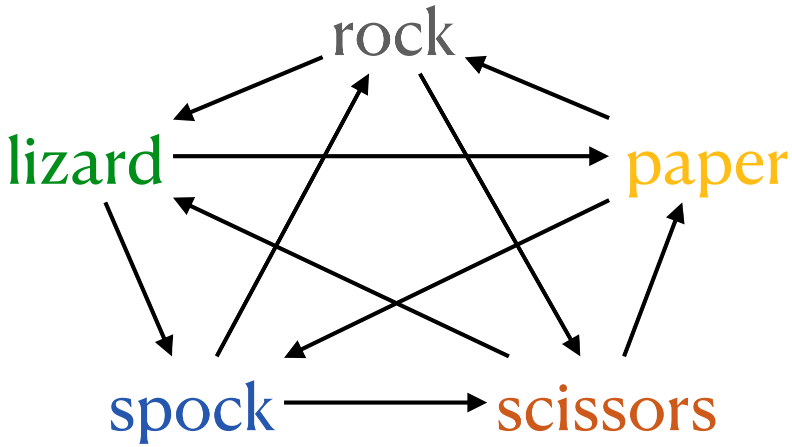
"*equality*" $= \subseteq \mathbb{N} \times \mathbb{N}$ is another a 2-ary relation

Notation

- A 2-ary relation is simply called ***binary relation***;
- for binary relations we use *infix notation* (e.g. $2 \leq 3$ or $a\ R\ b$);
- we use the notation $R(a_1, \ldots, a_k)$ to mean $(a_1, \ldots, a_k) \in R$.

Example: Rock-Paper-Scissors-Spock-Lizard*

The TV series *The Big Bang Theory* used an advanced version of the popular children's game Rock-Paper-Scissors, with two extra choices



Scissors cuts Paper covers Rock crushes Lizard poisons Spock smashes Scissors decapitates Lizard eats Paper disproves Spock vaporises Rock crushes Scissors

Question: how it is defined the relation "Beats"?

$$\text{Beats} = \left\{ \begin{array}{l} (\text{rock, scissor}), (\text{rock, lizard}), (\text{paper, rock}), (\text{paper, spock}), (\text{scissor, paper}), \\ (\text{scissor, lizard}), (\text{spock, scissor}), (\text{spock, rock}), (\text{lizard, paper}), (\text{lizard, spock}) \end{array} \right\}$$

(*) http://www.samkass.com/theories/RPSSL.html

Special types of binary relations

We will often be interested in relations $R \subseteq A \times A$ having the following properties

- R is ***reflexive*** if, for every $a \in A, \;\; a \; R \; a$;
- R is ***symmetric*** if, for every $a, b \in A, \; a \; R \; b$ implies $b \; R \; a$;
- R is ***antisymmetric*** if, for every $a, b \in A, \; a \; R \; b$ and $b \; R \; a$ implies $a = b$;
- R is ***transitive*** if,
  for every $a, b, c \in A, \; a \; R \; b$ and $b \; R \; c$ implies $a \; R \; c$;
- R is a ***pre-order relation*** if it is reflexive, antisymmetric, and transitive.
- R is an ***equivalence relation*** if it is reflexive, symmetric, and transitive.

Functions

- A ***function*** (or ***mapping***) f describes an input-output assignment
- the set of possible inputs to the function is called ***domain of*** f
- the outputs are the elements of a set called ***range of*** f

Notation

denotes that f is a function

$$\underbrace{f\colon}_{\text{denotes that } f \text{ is a function}} \overbrace{A}^{\text{domain}} \to \overbrace{B}^{\text{range}}$$

$$f(\overbrace{a}^{\text{input}}) = \overbrace{b}^{\text{output}}$$

Example: consider the function $f\colon \{0,1,2\} \to \{0,1,2\}$

n	$f(n)$
0	1
1	2
2	0

tabular definition

input-output expression definition

$$f(n) = n \bmod 2$$

Functions (continued)

inputs are k-tuples $(a_1, \ldots, a_k)$

- A ***k-ary function*** is a function of type $f\colon A_1 \times \ldots \times A_k \to B$; we call the $a_i \in A_i$ the ***arguments*** of f;
- k is called ***arity*** of f;
- if $k = 1$ the function is called ***unary***; if $k = 2$ it is called ***binary***;

Example: let the function $f\colon \{0,1,2\} \times \{0,1,2\} \to \{0,1,2\}$

f	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

tabular definition

input-output expression definition

$$f(n, m) = (n + m) \bmod 2$$

Functions are just relations!

- technically speaking, a function $f\colon A \to B$ is just a binary relation between A, B such that,
  for every element $a \in A$ there *exists a unique* element $f(a) \in B$.

- The corresponding relation is called ***the graph of f***

$$\mathcal{G}_f = \{(a, f(a)) \mid a \in A\} \subseteq A \times B$$

- when some elements of the domain don't have a corresponding element in the codomain we say that the function is ***partial***;
  we denote this by writing $f\colon A \rightharpoonup B$

Predicates (or Properties)

- A ***predicate*** (or ***property***) is a function with range $\{\text{true}, \text{false}\}$;

  Example: consider $\text{even}: \mathbb{N} \to \{\text{true}, \text{false}\}$ as the function mapping even numbers to true and odd numbers to false

$$\text{even}(n) = \begin{cases} \text{true} & \text{if } n = 2m \text{ for some } m \in \mathbb{N} \\ \text{false} & \text{otherwise}\,. \end{cases}$$

- a predicate is *equivalently described as a set* (the subset of inputs with value true).

  Example: the property even above is equivalent to the set

$$\text{even} = \{n \mid n = 2m \text{ for some } m \in \mathbb{N}\}\,.$$

Propositional Boolean Logic

The formulas of propositional Boolean logic are built from a set propositional variables (eg. $P, Q, R, \ldots$) and logical connectives

negation

$$\neg 0 = 1$$
$$\neg 1 = 0$$

conjunction

$$0 \wedge 0 = 0$$
$$0 \wedge 1 = 0$$
$$1 \wedge 0 = 0$$
$$1 \wedge 1 = 1$$

disjunction

$$0 \vee 0 = 0$$
$$0 \vee 1 = 1$$
$$1 \vee 0 = 1$$
$$1 \vee 1 = 1$$

implication

$$P \rightarrow Q = \neg P \vee Q$$

false

$$0 = P \wedge \neg P$$

true

$$1 = \neg 0$$

double implication

$$P \leftrightarrow Q = (P \rightarrow Q) \wedge (Q \rightarrow P)$$

exclusive or (XOR)

$$P \oplus Q = \neg (P \leftrightarrow Q)$$

Boolean laws

LAW	conjunction	disjunction
identity	$1 \wedge P = P$	$0 \vee P = P$
null element	$0 \wedge P = 0$	$1 \vee P = 0$
idempotence	$P \wedge P = P$	$P \vee P = P$
inverse	$P \wedge \neg P = 0$	$P \vee \neg P = 0$
commutativity	$P \wedge Q = Q \wedge P$	$P \vee Q = Q \vee P$
associativity	$P \wedge (Q \wedge R) = (P \wedge Q) \wedge R$	$P \vee (Q \vee R) = (P \vee Q) \vee R$
distributivity	$P \wedge (Q \vee R) = (P \wedge Q) \vee (P \wedge R)$	$P \vee (Q \wedge R) = (P \vee Q) \wedge (P \vee R)$
absorption	$P \wedge (P \vee Q) = P$	$P \vee (P \wedge Q) = P$
DeMorgan	$\neg(P \wedge Q) = \neg P \vee \neg Q$	$\neg(P \vee Q) = \neg P \wedge \neg Q$
double negation	$\neg\neg P = P$	

Logical notation

- the logical connectives from Boolean logic are also the standardly used symbols for writing mathematical statements
- we will often use quantifiers (from *First Order Logic*)
  - **universal quantifiers** (∀) are used to mean "for all"
    Example: $\forall n \in \mathbb{N} \,.\, n \leq n + 1$ ("for all natural numbers n, the number n is less than or equal to its successor")
  - **existential quantifiers** (∃) are used to mean "there exists"
    Example: $\exists n \in \mathbb{Z} \,.\, 2n \leq n$ ("there exists an integer n whose double $2n$ is less than or equal to n")

Types of proofs

Definition: A ***proof*** is a convincing logical argument that shows that a mathematical statement is true

Proof by construction

Used to show that an object exists by showing how to construct it.
This is the best way to attempt a proof because
it is *direct* and the *easiest* proof method available to us

Proposition

If n and m are consecutive integers, then $n + m$ is odd.

proof

Assume n and m are consecutive integers. Because they are consecutive, we know that $m = n + 1$. Thus, the sum $n + m$ can be rewritten as $2n + 1$. Thus, there exists a number k such that $n + m = 2k + 1$, so the sum is odd.

Proof by contradiction

In an argument by contradiction, one proves a statement by first assuming that the statement is false and than showing that this assumption leads to a false consequence (called *contradiction*)

Proposition

There are no integers n and m for which $12n + 6m = 2$.

proof

Assume, for the sake of contradiction, that there exists $n, m \in \mathbb{Z}$ such that $12n + 6m = 2$. By dividing by 6 on both sides of the equation, we obtain

$$2n + m = \frac{1}{3}.$$

This is a contradiction, since by closure properties $2n + m$ is an integer but $\frac{1}{3}$ is not.

Therefore, it must be that no integers n and m exist for which $12n + 6m = 2$.

Proof by induction

It is an advanced method used to prove that all the elements of an infinite set (e.g., the natural numbers) have a specific property.

Induction on natural numbers:

Let P be a property on $\mathbb{N}$. We want to prove $P(n)$ for all $n \in \mathbb{N}$ (i.e., we shall prove $P(0), P(1), P(2), \ldots$).

- the ***base case*** (or ***basis***) proves that $P(0)$ is true.
- the ***inductive step*** proves that, for each $i \geq 0$,
  if $P(i)$ is true (*inductive hypothesis*), then so is $P(i+1)$.

induction proof principle on $\mathbb{N}$

$$\big(P(0) \wedge \forall i \geq 0.(P(i) \Rightarrow P(i+1))\big) \Rightarrow \forall n \in \mathbb{N} \,.\, P(n)$$

proof by induction (Example)

Proposition

For all integers $n \geq 0$, we have $(n+1)^2 = 1 + \sum_{k=1}^{n} (2k+1)$.

proof

Let $P(n)$ be the statement $(n+1)^2 = 1 + \sum_{k=1}^{n} (2k+1)$. We proceed by induction.

Base case: We prove $P(0)$. When $n = 0$, the equation above is $(0+1)^2 = 1 + 0$.

Inductive step: For $i \geq 0$, assume $P(i)$ is true. Next we show $P(i+1)$ is also true:

$$
\begin{aligned}
1 + \sum_{k=1}^{i+1} (2k+1) &= \left(1 + \sum_{k=1}^{i} (2k+1)\right) + \left(2(i+1)+1\right) \\
&= (i+1)^2 + 2(i+1) + 1 \qquad \text{(inductive hypothesis)} \\
&= i^2 + 4i + 4 \\
&= (i+2)^2 \\
&= ((i+1)+1)^2 .
\end{aligned}
$$

variations ...

induction proof principle on $\mathbb{N}$

$$\big(P(0) \land \forall i \geq 0.(P(i) \Rightarrow P(i+1))\big) \Rightarrow \forall n \in \mathbb{N}\,.\,P(n)$$

once we understand how it works, it is simple to see that other variations to the same principle are also valid

on a different base case

$$(P(b) \land \forall i \geq b\,.\,(P(i) \Rightarrow P(i+1))) \Rightarrow \forall n \geq b\,.\,P(n)$$

complete (or strong) induction

$$\big(P(0) \land \forall i \geq 0.(\forall j \leq i\,.\,P(j)) \Rightarrow P(i+1)\big) \Rightarrow \forall n \in \mathbb{N}\,.\,P(n)$$

complete induction (Example)

Complete induction is very convenient in certain cases...

Fibonacci sequence

$$\mathrm{fib}_0 = 0$$
$$\mathrm{fib}_1 = 1$$
$$\mathrm{fib}_n = \mathrm{fib}_{n-1} + \mathrm{fib}_{n-2}$$

Proposition

golden ration

For all $n \geq 0$, $\mathrm{fib}_n = \dfrac{\phi^n - \psi^n}{\phi - \psi}$, where $\phi = \dfrac{1+\sqrt{5}}{2}$ and $\psi = \dfrac{1-\sqrt{5}}{2}$.

proof

Let $P(n)$ be the statement $\mathrm{fib}_n = \dfrac{\phi^n - \psi^n}{\phi - \psi}$. We proceed by complete induction.

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proof

Base case: We prove that $P(0)$ and $P(1)$ are true.
In the first case, $\mathrm{fib}_0 = 0 = \frac{1-1}{\phi-\psi}$; as for the second case, $\mathrm{fib}_1 = 1 = \frac{\phi-\psi}{\phi-\psi}$.

Inductive step: For each $i \geq 1$, we assume that $P(j)$ is true for all $j \leq i$, and use this to prove $P(i+1)$, as follows:

$$\begin{aligned}
\mathrm{fib}_{i+1} = \mathrm{fib}_i + \mathrm{fib}_{i-1} &= \frac{\phi^i - \psi^i}{\phi-\psi} + \frac{\phi^{i-1} - \psi^{i-1}}{\phi-\psi} && \text{(ind. hp)}\\
&= \frac{(\phi^i + \phi^{i-1}) - (\psi^i + \psi^{i-1})}{\phi-\psi}\\
&= \frac{(\phi^{i+1} - \psi^{i+1})}{\phi-\psi}
\end{aligned}$$

The last step follows from the fact that ϕ and ψ are the roots of the polynomial $x^2 - x - 1$, thus $\phi^2 - \phi - 1 = 0$ and $\psi^2 - \psi - 1 = 0$.
Therefore, $\phi^2 = \phi + 1$ and by multiplying both sides by ϕ^{i-1} we get $\phi^{i+1} = \phi^i + \phi^{i-1}$. The same argument can be used to show $\psi^{i+1} = \psi^i + \psi^{i-1}$.

Well-founded sets

Induction does not work only on natural numbers,
but more in general on well-funded sets!

Let $\sqsubset$ be a binary relation on a set X.

- $\sqsubset$ is a ***well-founded relation*** if every nonempty subset $Y \subseteq X$ has a minimal element:

$$\forall Y \subseteq X \,.\, \big(Y \neq \varnothing \Rightarrow \exists m \in Y \,.\, \forall y \in Y \,.\, \neg(y \sqsubset m)\big)$$

- In this case $(X, \sqsubset)$ is called well-founded set.

Note

By definition, X has no infinite decreasing sequence

$$x_0 \sqsupset x_1 \sqsupset x_2 \sqsupset x_3 \sqsupset x_4 \sqsupset \ldots$$

Examples of well-founded sets

- The set of natural numbers $\mathbb{N}$ with "less than" relation $<$;
- The set of tree structures with the "proper subtree" relation
- The set of tuples over $\mathbb{N}$, with strict "lexicographic" order
- The set of algebraic expressions, with "subexpression" relation
- $\mathbb{N} \times \mathbb{N}$ ordered by $(n_1, n_2) \sqsubset (m_1, m_2)$ iff $(n_1 < m_1) \wedge (n_2 < m_2)$
- and so on...

Well-founded induction

Let $(X, \sqsubset)$ be a well-funded set and P a property on X.

Induction on well-founded sets:

To prove that $P(x)$ is true for all $x \in X$, we just need to

- the ***base case*** proves $P(m)$ for all minimal elements of X.
- the ***inductive step*** proves that,
  for each $i \in X$, if $P(j)$ is true for all $j \sqsubset i$, then so is $P(i)$.

well-founded induction proof principle

$$\big(\forall i \in X \,.\, (\forall j \in X \,.\, j \sqsubset i \Rightarrow P(i)) \Rightarrow P(i)\big) \Rightarrow \forall x \in X \,.\, P(x)$$

well-founded induction (Example)

Proposition

A binary tree T of height $h(T)$ has at most $2^{h(T)}$ leaves.

proof

Let $\mathbb{T}$ be the set of binary trees and $P(T)$ be the statement $|\text{leaves}(T)| \leq 2^{h(T)}$, where $T \in \mathbb{T}$ and leaves(T) denotes the set of leaves of T. The relation $T < T'$ iff T is a *proper subtree* of T' is well-founded. Therefore, we can proceed by well-founded induction on $(\mathbb{T}, <)$.

Base case: We prove $P(E)$ for the empty-tree E. The empty tree has no leaves and its height is 0 (in symbols, $\text{leaves}(E) = \varnothing$ and $h(E) = 0$). Thus, $P(E)$ is just $0 \leq 2^0$.

Inductive step: For all $T \in \mathbb{T}$, assume that $P(S)$ is true whenever $S < T$. Next we show that $P(T)$ is also true. Let T_L and T_R the left and right subtree of T.

$$\begin{aligned}
|\text{leaves}(T)| &= |\text{leaves}(T_L)| + |\text{leaves}(T_R)| \\
&\leq 2^{h(T_L)} + 2^{h(T_R)} && \text{(inductive hypothesis)} \\
&\leq 2 \cdot 2^{\max\{h(T_L), h(T_R)\}} \\
&= 2^{h(T)}
\end{aligned}$$